

GLS UNIVERSITY — DEPARTMENT OF COMPUTER APPLICATIONS

Course: Bachelor of Computer Applications (BCA) | Semester 3 Syllabus | Academic Year: 2025

BCA Semester 3 Course Structure & Credit Table

Course Code	Subject Name	Course Type	Credits	Theory Marks	Practical Marks
BCA-301	Database Management Systems (DBMS)	Core Theory	4	70	30
BCA-302	Object Oriented Programming with Java	Core Theory	4	70	30
BCA-303	Data Structures & Algorithms	Core Theory	4	70	30
BCA-304	Web Application Development (RAG/Python)	Core Theory	4	70	30
BCA-305	DBMS & Java Lab	Practical Lab	2	0	50

Detailed Syllabus Units for BCA-301: Database Management Systems

- Unit 1: Introduction to Relational Databases, ER Diagrams, Relational Algebra.
- Unit 2: SQL Fundamentals, DDL, DML, Complex Queries, Joins, and Subqueries.
- Unit 3: Database Normalization (1NF, 2NF, 3NF, BCNF), Functional Dependencies.
- Unit 4: Transaction Management, ACID Properties, Concurrency Control, Recovery.